Fine-Grained Image Classification

Computer Vision Report

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1 ABSTRACT

This work is developed in the context of a competitive machine learning challenge on fine-grained image classification (FGIC), emphasizing predictive performance, methodological rigor, and generalization under limited labeled data.

We compare Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), focusing on ViT-based models due to their superior competition performance. Experiments span multiple fine-grained datasets—flowers, aircraft, and cars—evaluated via standard top-1 accuracy on a held-out test set.

CNN-based approaches (ResNet50, ResNet152) suffered from overfitting and poor generalization on our constrained dataset. ViT-based models consistently outperformed them: two configurations achieved approximately 72% accuracy on the competition mammal dataset. Our findings highlight the decisive impact of architectural choice, augmentation strategy, and learning rate scheduling on fine-grained classification performance.

2 INTRODUCTION

Fine-Grained Image Classification (FGIC) is a challenging subfield of computer vision that focuses on distinguishing between visually similar subcategories within a broader semantic class. Unlike coarse-grained classification—where categories differ by clearly identifiable visual traits—FGIC requires the recognition of subtle, localized visual cues under significant variations in pose, illumination, background, and scale. The core difficulty arises from the combination of small inter-class variance and large intra-class variance: models must capture fine visual details while remaining robust to appearance changes. Applications span wildlife monitoring, biodiversity assessment, and biological research, where species-level visual distinctions are often minimal.

Recent advances in deep learning have significantly improved fine-grained recognition. Transformer-based architectures have shown strong potential due to their ability to model long-range dependencies through attention mechanisms. Vision Transformers (ViTs) have emerged as a competitive alternative to convolutional neural networks, especially in scenarios where capturing global structure and subtle inter-region relationships is critical.

In this work, we address fine-grained mammal image classification with an end-to-end pipeline integrating data preparation, augmentation, representation learning, and classification in a unified framework. The approach is designed for limited-data regimes, emphasizing robustness and generalization. We compare CNN baselines against ViT-based models, investigate a ViT+SpinalNet hybrid architecture, and provide an empirical analysis of how architectural and training choices impact performance.

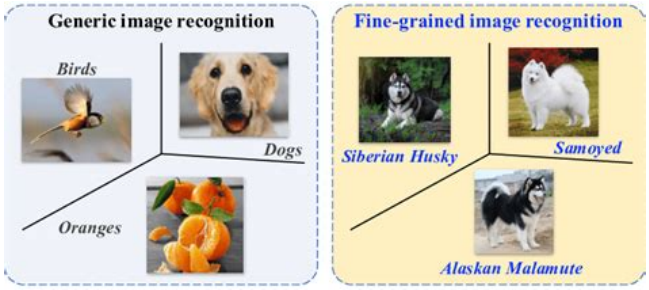

Figure 1: Distinction between generic image recognition (left) and fine-grained classification (right), where subcategories are visually near-identical.

3 RELATED WORK

In this section, we provide a concise overview of the core techniques underlying our approach.

3.1 Approaches to Fine-Grained Image Recognition

Fine-grained image recognition methods broadly fall into three categories [10]:

- **Localization-Classification Subnetworks:** Methods that detect or segment discriminative object parts before classification—using CNNs with attention mechanisms, Spatial Transformer Networks (STNs), or part-based detectors—to focus on semantically meaningful regions.
- **End-to-End Feature Encoding:** Pipelines that learn discriminative representations directly through deep networks without explicit part detection, reducing dependence on fine-grained annotations.
- **Recognition with External Information:** Approaches that incorporate supplementary data or annotations to aid distinction between visually similar categories.

3.2 Vision Transformers

Transformers, introduced by Vaswani et al. [9], rely on self-attention to capture long-range dependencies in sequences. The Vision Transformer (ViT) [4] adapts this to images by treating them as sequences of fixed-size patches (e.g., 16×16 pixels). Each patch is flattened, linearly projected into an embedding, and enriched with a positional encoding; the resulting sequence is processed through standard Transformer encoder layers. A learnable `[CLS]` token is prepended, and its final state serves as the image representation for classification.

Unlike CNNs, ViT has a global receptive field from the first layer, making it particularly suited for fine-grained tasks where discriminative cues may span the entire image rather than being localized.

3.3 Data Augmentation

Data augmentation artificially expands the training set via label-preserving transformations, combating overfitting in limited-data settings [8]. We used the `torchvision.transforms.v2` library with the following pipeline applied to the training set only:

```
def get_transforms():
    return v2.Compose([
        v2.Pad(10),
        v2.Resize(256),
        v2.RandomCrop(size=(224, 224)),
        v2.RandomHorizontalFlip(p=0.5),
        v2.RandomRotation(15),
        v2.ColorJitter(brightness=0.2,
                       contrast=0.2,
                       saturation=0.2),
        v2.ToTensor(),
        v2.Normalize(mean=[0.485, 0.456, 0.406],
                     std=[0.229, 0.224, 0.225])])
```

Geometric transforms (padding, random crop, flip, rotation) address pose and scale variation; color jitter handles varying lighting. Normalization aligns inputs with the ImageNet statistics expected by pretrained models.

3.4 Training Components

3.4.1 Activation Functions. We tested ReLU, LeakyReLU, PReLU, Swish, GELU, and Sigmoid [2, 7]. Empirical evaluation on the Cars dataset identified **LeakyReLU** as the best-performing activation, achieving 88.5% validation accuracy versus 87.0% for ReLU. It was adopted in the final ViT+SpinalNet model.

3.4.2 Learning Rate Scheduling. We used a **warmup + cosine annealing** scheduler via the `get_cosine_schedule_with_warmup` function from the HuggingFace *transformers* library. During warmup, the learning rate increases linearly from near zero to a peak value, preventing unstable weight updates in the early training steps. After warmup, it decays following a cosine curve, allowing fine-grained convergence and helping the model escape local minima.

3.4.3 AdamW Optimizer. **AdamW** [6] extends Adam by decoupling weight decay from the gradient update, resulting in more effective regularization. This separation prevents weight decay from interfering with the adaptive moment estimates, making AdamW particularly well-suited for transformer architectures. T

3.5 SpinalNet

SpinalNet [5] is inspired by the hierarchical, segmented processing of the human spinal cord. Rather than routing all features through a single fully connected layer, SpinalNet splits the input into segments and processes them progressively through multiple *spinal layers*, with each layer receiving the current segment and the accumulated output of all previous layers. This design improves gradient flow, mitigates the vanishing gradient problem in deep networks, and provides implicit regularization through structural sparsity.

4 METHODOLOGY

This section describes the three architectural approaches explored, in order of development. Each step was motivated by the limitations of the previous one.

4.1 CNN Baseline: ResNet

We began with pretrained ResNet50 and ResNet152 models as CNN baselines. Both were fine-tuned on our datasets using two key adaptations.

4.1.1 Freezing Strategy. To avoid catastrophic forgetting when fine-tuning on a small dataset, we froze the convolutional backbone and only trained the final classifier layers:

```
for name, child in model.named_children():
    if isinstance(child, nn.Conv2d):
        for param in child.parameters():
            param.requires_grad = False
   elif name in ['layer1','layer2','layer3','layer4']:
        for param in child.parameters():
            param.requires_grad = False
```

This retains the low-level features learned on ImageNet while allowing the head to adapt to the task.

4.1.2 Custom Classifier Head. We replaced ResNet's default fully connected head with a deeper classifier to improve feature discrimination:

```
model.fc = nn.Sequential(
    nn.Linear(num_features, 256),
    nn.ReLU(inplace=True),
    nn.Dropout(0.1),
    nn.Linear(256, num_classes)
)
```

4.1.3 Outcome and Failure Analysis. Despite these adaptations, ResNet performed poorly. ResNet152 achieved 56.32% and ResNet50 only 34.70% on the Flowers dataset. ResNet50 showed pronounced overfitting, suggesting its lower representational capacity was insufficient for the subtle inter-class differences in FGIC tasks. The frozen backbone, while preventing catastrophic forgetting, also limited the model's ability to learn task-specific features. These results motivated a full architectural shift away from CNNs.

4.2 Vision Transformer (ViT)

Switching to ViT produced immediate, dramatic improvements—from under 57% to over 93% on the Flowers dataset. We used the `vit_large_patch16_224` model from the *timm* library [3]: a large ViT pretrained on ImageNet-21k, processing images as sequences

of 16×16 pixel patches. The classification head was replaced with a linear layer matching the target number of classes:

Code 1 Model Creation

```
model_ft = timm.create_model(
    'vit_large_patch16_224',
    pretrained=True,
    num_classes=Num_class)
model_ft = model_ft.to(device)
```

ViT's global self-attention mechanism captures inter-patch dependencies across the full image in every layer—unlike CNNs, which build receptive fields gradually through pooling. This global view is precisely what fine-grained discrimination requires: subtle cues (e.g., fur texture, body proportions) often span distant image regions and are invisible to locally-constrained CNN features.

4.2.1 Optimizer and Scheduler. All ViT models were trained with:

- **Optimizer:** AdamW (`lr=0.001`, `weight_decay=0.01`), chosen for its superior regularization over standard Adam on transformer architectures.
- **Scheduler:** Warmup + cosine annealing (500 warmup steps over 15 training epochs), stabilizing early training and enabling precise fine-tuning convergence.
- **Loss:** `LabelSmoothingCrossEntropy`, which prevents overconfident predictions and improves generalization.
- **Training aids:** Early stopping on validation loss, checkpointing at every epoch.

Code 2 Optimizer and Scheduler Setup

```
optimizer_ft = torch.optim.AdamW(
    model_ft.parameters(),
    lr=0.001, weight_decay=0.01)
num_warmup_steps = 500
num_training_steps = 15 * len(dataloaders['train'])
scheduler_ft = get_cosine_schedule_with_warmup(
    optimizer_ft,
    num_warmup_steps=num_warmup_steps,
    num_training_steps=num_training_steps)
```

4.2.2 Three Model Variants. Three ViT configurations were trained:

- **`vit_slow`:** batch size 12, `lr=0.001`.
- **`vit_fast`:** batch size 32, `lr=0.003`.
- **`vit_WITH_spiralnet`:** same base configuration as `vit_slow` but with the SpinalNet classifier head (described below).

4.3 ViT with SpinalNet Classification Head

The third model replaces ViT's default linear head with a **SpinalNet** classifier. The motivation was to improve feature integration at the classification stage: rather than mapping the `[CLS]` token embedding directly to class logits, SpinalNet processes the representation through progressively combined segments, each layer receiving a new input segment plus the concatenated output of all prior layers.

Code 3 SpinalNet Head Attachment

```
class SpinalNet(nn.Module):
    def __init__(self): ...
    def forward(self, x): ...
model_ft.head = SpinalNet()
```

Activation function: LeakyReLU was selected over ReLU after a systematic sweep across six activation functions on the Cars dataset. LeakyReLU allows small gradients for negative inputs, preventing dead neurons and providing a consistent 1–2% accuracy advantage.

Additional training elements in this model (not present in `vit_fast`/`vit_slow`):

- **Early Stopping:** Training halts when validation loss fails to improve, preventing wasted compute and overfitting.
- **Label Smoothing Cross-Entropy:** Replaces the standard cross-entropy loss used by the two baseline ViT models, further regularizing training.

4.3.1 Architectural Comparison. The three models differ primarily at the classification head:

- **`vit_slow`/`vit_fast`:** `nn.Linear(in_features, num_classes)`—simple, direct mapping from `[CLS]` token to logits.
- **`vit_WITH_spiralnet`:** SpinalNet head with multiple spinal layers and LeakyReLU activations—hierarchical feature integration before final classification.

The results (72% vs. 71.8%) show that the SpinalNet head performs comparably at this scale; the added architectural complexity improves gradient flow but does not yield a decisive accuracy gain on the available data. The architectural value of SpinalNet may become more apparent with larger or more diverse fine-grained datasets.

5 EXPERIMENTS

We conducted 43 experiments across three datasets to progressively refine our approach. Here, we report the key decisions and lessons learned.

5.1 Phase 1 — Flowers Dataset (Baseline Establishment)

The Oxford Flowers 102 dataset (8,189 images, 102 classes) served as our initial benchmark. We began with CNN baselines (ResNet152, ResNet50) before transitioning to ViT.

Key results: ResNet152 achieved 56.32% and ResNet50 only 34.70%, with the latter showing clear signs of overfitting. These results confirmed that CNN-based fine-tuning was inadequate for the subtle inter-class differences in FGIC under limited data. Switching to `vit_large_patch16_224` produced a dramatic jump to 93.43% (without background classes) and 98.25% (with background classes), firmly establishing ViT as the architecture of choice.

5.2 Phase 2 — Aircraft Dataset (Hyperparameter Refinement)

The FGVC Aircraft dataset (10,000+ images, 100 classes) served as the main experimental testbed, with 27 iterative experiments.

Primary challenge: overfitting. Early models rapidly overfit—Experiment 2 reached 97.5% training accuracy in the first epoch but only 74.8% validation accuracy. This drove extensive exploration of regularization strategies.

Key lessons learned:

- **Augmentation balance:** Aggressive augmentation (CutMix, MixUp, RandAugment combined) proved too strong, collapsing training accuracy to 51%. Reverting to geometric-only `v2.transforms` augmentation (Exp. 13, 82.51%) struck the right balance.
- **LR scheduling:** Moving from `StepLR` to `ReduceLROnPlateau` (Exp. 12, 79.36%) and then adding manual `v2.transforms` augmentation (Exp. 13, 82.51%) produced the best results on this dataset.
- **Label smoothing:** Adding `LabelSmoothing` (Exp. 11 onward) consistently helped generalization.
- **Reproducibility:** Results varied significantly without a fixed random seed (Exp. 6), motivating the use of `torch.manual_seed` from Exp. 7 onward.
- **Activation functions:** Switching from ReLU to LeakyReLU (Exp. 27, 79.65%) showed a consistent improvement, informing the final model.

5.3 Phase 3 — Cars Dataset (Final Configuration)

The Stanford Cars dataset [1] (16,185 images, 196 classes) was used to finalize training settings. We fixed the warmup + cosine annealing scheduler and AdamW optimizer, and ran a systematic activation function sweep. **Key findings:**

- **LeakyReLU** achieved the best validation accuracy (88.5%), confirming the finding from Phase 2. PReLU, GELU, and Sigmoid performed significantly worse (69–75%).
- **Learning rate sensitivity:** Tripling the LR from the best baseline (Exp. 10, 94.60%) and then halving it (Exp. 11, 95.46%) demonstrated that conservative LR tuning around a strong baseline is highly effective.
- **Batch size and weight decay:** Increasing batch size or reducing weight decay yielded suboptimal results, reinforcing the final configuration of batch=32 and weight_decay=0.01.

These findings directly informed the final competition model: `vit_large_patch16_224` with AdamW, warmup + cosine annealing scheduling, LeakyReLU (in the SpinalNet variant), and `LabelSmoothingCrossEntropy`.

6 RESULTS

Final competition performance was evaluated by running the trained models on the unseen test set and submitting the predictions to the competition system, where they were compared against the ground-truth labels.

Model performance:

- **`vit_fast`**: This model was fully trained and tested internally, but due to a submission issue it could not be uploaded to the competition platform.
- **`vit_slow`**: Achieved the best result with a test accuracy of **72.0%**.
- **`vit_WITH_spiralnet`**: Reached a very similar performance with a final accuracy of **71.8%**.

The three models shared a very similar overall architecture, with the main differences being the training configuration and the addition of the SpinalNet head in the third model. The `vit_slow` model was trained using a smaller batch size (12) and a lower learning rate (0.001), while `vit_fast` used a batch size of 32 and a learning rate of 0.003. Despite the extra complexity introduced by the SpinalNet head, the `vit_WITH_spiralnet` model performed almost identically to the best-performing model, finishing only 0.2% lower. This indicates that the SpinalNet head was able to capture additional hierarchical feature relationships without reducing overall performance.

Regarding overfitting, the training and validation accuracies remained very close throughout training, with validation accuracy occasionally even exceeding training accuracy. This suggests that the regularization methods were effective and that the models generalized well to unseen data. Techniques such as data augmentation, label smoothing, AdamW weight decay, and early stopping all contributed to more stable training and helped prevent overfitting. The `vit_WITH_spiralnet` model also showed slightly slower learning during the early epochs, which was expected because of its more complex classification head.

Overall, the strongest result was achieved by `vit_slow`, with a final test accuracy of **72%**. Although there is still room for improvement, the results demonstrate that the proposed approach is already capable of reasonably reliable fine-grained mammal species classification.

7 CONCLUSION

This report explored two families of architectures for fine-grained image classification: CNNs (ResNet50, ResNet152) and Vision Transformers (ViT). The CNN baselines failed to generalize under limited data, confirming that their locally-constrained feature extraction is poorly suited to FGIC tasks where discriminative cues are subtle and globally distributed. Vision Transformers, by contrast, leverage self-attention to capture inter-patch dependencies across the full image, achieving far superior results on every dataset tested—most decisively, jumping from 56% (ResNet152) to 93% (ViT) on the Flowers dataset.

Among the ViT variants, the standard linear head (`vit_slow`) and the SpinalNet-augmented head (`vit_WITH_spiralnet`) produced near-identical competition accuracy (72% vs. 71.8%), suggesting that the hierarchical structure of SpinalNet adds stability and gradient flow benefits without decisively improving discriminative power at this data scale.

The primary challenges of this work were managing a small and format-diverse dataset, selecting appropriate regularization strategies, and tuning the learning rate schedule. The most impactful decisions were the architectural switch from CNN to ViT, the adoption of the warmup + cosine annealing learning rate schedule, and the use of AdamW with label smoothing.

Limitations include the restricted set of datasets and hyperparameter configurations explored, and the inability to test all activation function and optimizer combinations. Future work could investigate

larger ViT variants, more sophisticated part-based attention, or self-supervised pretraining on domain-specific data to push accuracy beyond the current 72% ceiling.

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